SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717- SOFTWARE TESTING

EXPERIMENT 7

Case Study: Food Delivery Application – Negative Test Scenarios and Test Cases

Aim:

To design and execute negative test scenarios and test cases for a Food Delivery Application to verify that invalid orders, unavailable services, and failed transactions are handled properly.

Problem Description

A food delivery application allows customers to register, log in, search restaurants and food items, add items to a cart, enter a delivery address, and place orders. The system must validate item availability, quantity, address, delivery coverage, and payment before confirming an order. The testing team is responsible for verifying whether the application works correctly under valid and invalid conditions. They must prepare positive and negative test cases and document any defects identified during testing.

Functional Requirements

1. Users must register/login with valid details.

2. Restaurants and menu items must be searchable.

3. Only available food items can be ordered.

4. Quantity must be greater than zero and within available limits.

5. A valid delivery address is required.

6. Order confirmation requires successful payment.

7. Invalid inputs must show appropriate error messages.

3. Negative Test Cases

Negative testing verifies how the application behaves when invalid, incomplete, or unexpected inputs are provided.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with wrong password	Correct email, wrong password	Login denied	Pass
NT02	Search with invalid/blank input	Submit blank search	Validation or appropriate result displayed	Pass
NT03	Order unavailable item	Select unavailable food item	Item cannot be ordered	Pass
NT04	Enter zero quantity	Set quantity to 0	Quantity rejected	Pass
NT05	Use invalid delivery PIN	Enter invalid/non-serviceable PIN	Delivery address rejected	Pass
NT06	Order outside delivery	Use address outside	Order should not be	Fail

	area	restaurant coverage	accepted	
NT07	Payment declined	Simulate failed payment	Order not confirmed	Pass
NT08	Submit empty address	Leave mandatory address fields blank	Required-field messages displayed	Pass

4. Detailed Execution of a Failed Test Case

Test Case NT06
Failed

Test scenario: Order outside delivery area

Precondition

A valid cart is available and the entered address is outside the restaurant's delivery area.

Test steps

1. Login.
2. Select an available food item.
3. Add it to cart.
4. Enter an address outside delivery coverage.
5. Proceed to place order.

Expected result

Application should reject the address and prevent order placement.

Actual result

Application accepts the address and allows the order to proceed.

Conclusion

The test fails because delivery-area validation is not enforced.

5. Sample Bug Report

A bug report is a document that records a defect discovered during software testing. It provides developers with sufficient information to reproduce, investigate, and fix the issue.

Field	Details
Bug ID	BR-001
Linked Test Case	NT06
Severity	High
Priority	High
Bug Title	Order outside delivery area
Environment	Application: Food Delivery Application; Browser: Google Chrome; Operating System: Windows 10; Test Module: Relevant application module
Preconditions	A valid cart is available and the entered address is outside the restaurant's delivery area.
Steps to Reproduce	1. Login. 2. Select an available food item. 3. Add it to cart. 4. Enter an address outside delivery coverage. 5. Proceed to place order.
Expected Result	Application should reject the address and prevent order placement.

Actual Result	Application accepts the address and allows the order to proceed.
Impact	The defect may cause incorrect system behavior, data inconsistency, transaction errors, or poor user experience.
Suggested Fix	Apply validation on the client and server side and repeat the validation at the final submission/transaction stage.
Status	Open — awaiting developer investigation and correction.

6. Bug Life Cycle

New — Tester reports the defect
Assigned — Developer receives the defect
In Progress — Developer investigates and fixes it
Fixed — Developer submits the correction
Retesting — Tester verifies the correction
Closed — Tester confirms the defect is resolved
If the defect still exists after the fix, the tester reopens the bug and sends it back for further correction.

7. Difference Between Positive and Negative Testing

Positive Testing	**Negative Testing**
Uses valid inputs.	Uses invalid or unexpected inputs.
Checks whether expected operations succeed.	Checks whether errors are handled correctly.
Example: valid login or valid transaction.	Example: wrong password or invalid transaction.
Expected result is successful execution.	Expected result is safe rejection or appropriate error handling.